POSTER · UNIT V

Flight modes at a glance

Each step up adds automation, and a sensor that can fail

MODE	IT HOLDS	NEEDS	PILOT STILL OWNS
Acro / Rate	**Body rotation rate**	Gyro	Everything else
Stabilize	**Level attitude**	+ accelerometer	Throttle and position
AltHold	**Altitude**	+ barometer	Horizontal position (drifts)
Loiter / PosHold	**Altitude and position**	+ GNSS + compass	Where to go
Guided / Auto	**Path to target or mission**	+ valid mission or GCS	Supervise, ready to override
RTL	**Climb, return, descend**	+ valid home + energy	Watch obstacles and wind
Land	**Controlled descent**	Baro (GNSS if available)	Keep the area clear

Rule 1. Lose a sensor, drop to the highest mode that does not need it. GNSS gone: AltHold. Baro bad: Stabilize.

Rule 2. In AltHold and Loiter, low stick = descend. In Stabilize, low stick = low motors.

Rule 3. RTL is only as safe as its home point, its altitude, its energy and its heading estimate.

Names follow ArduPilot Copter; PX4 equivalents: Manual, Altitude, Position, Mission, Return, Land